Local data

1D CNN + private FiLM

Federated averaging

House 1

House 2

House K

θ, ϕ_1

θ, ϕ_2

θ, ϕ_k

CNN weights c only

FedAvg

$\theta \leftarrow \frac{1}{k}\sum_{k=1}^{K} \theta_k$

50 rounds

Shared CNN θ

Private FiLM ϕ_k

Upload θ_k Broadkast θ

Local Training

The first 50 rounds: Update θ_k, ϕ_k

The last 30 rounds: Freeze θ_k, Update ϕ_k

θ_k, ϕ_k

Training target y_k

Training batch $\tilde{x}_k$

$h_k = \gamma_k \odot z_k + \beta_k$

$p_k(w)$

$e = p_k - y_k$

Huber loss
current δ_u

Backprop
+ Adam

Detach |e|

$Q_{0.75}$

EMA + bounds

δ_{u+1}